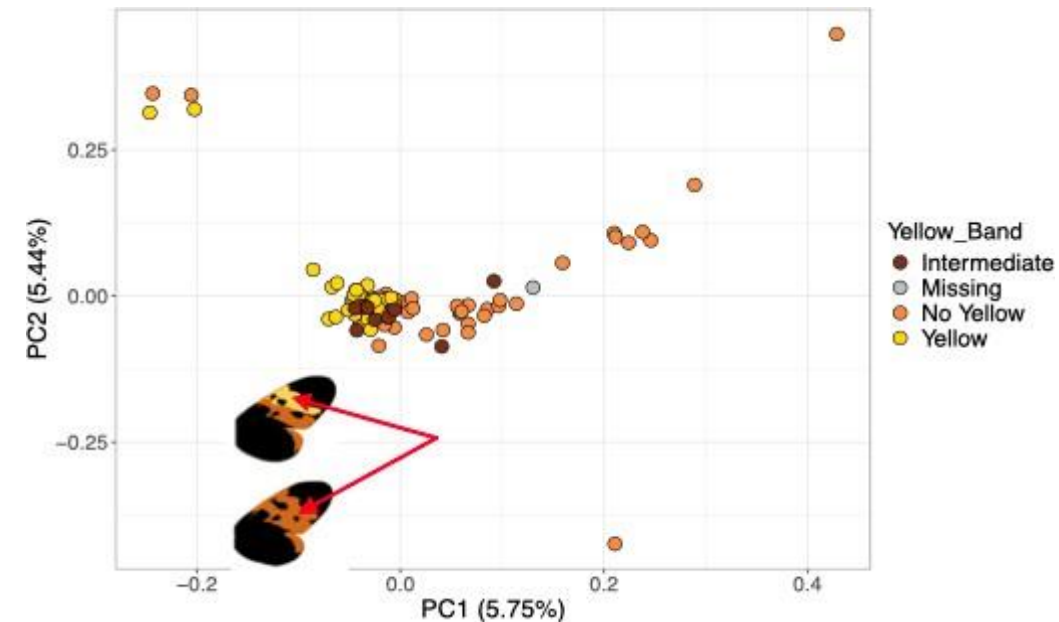


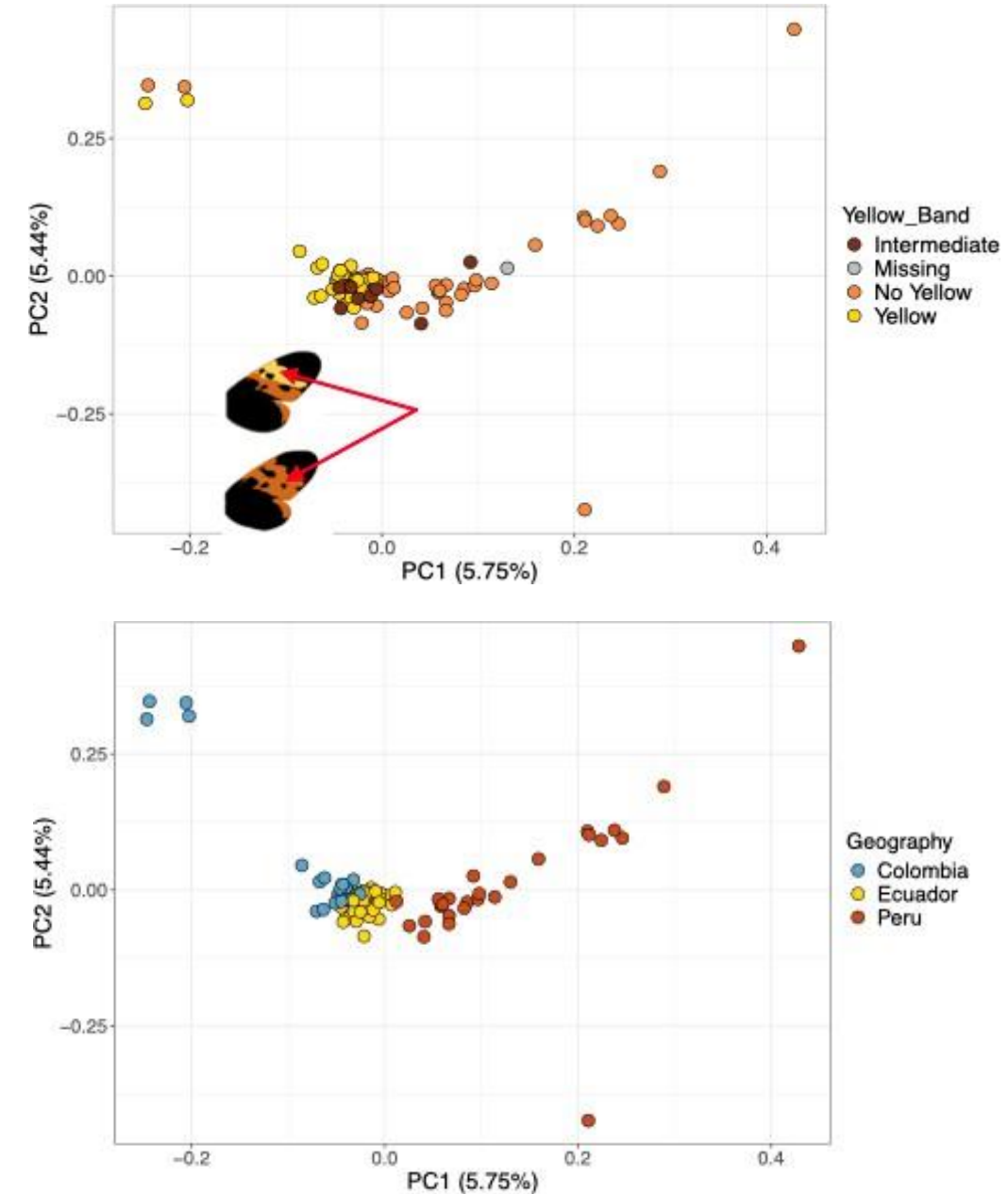
PCA – Principal Components Analysis

- **Complexity reduction:** PCA reduces the dimensionality of the data
- PCA identifies the **main axes of variation** in a dataset with each axis being independent of the next
- **Eigenvalues:** each axis has an eigenvalue, that can be transformed into the percentage of variance explained
- **Eigenvectors:** each individual has a position on each axis



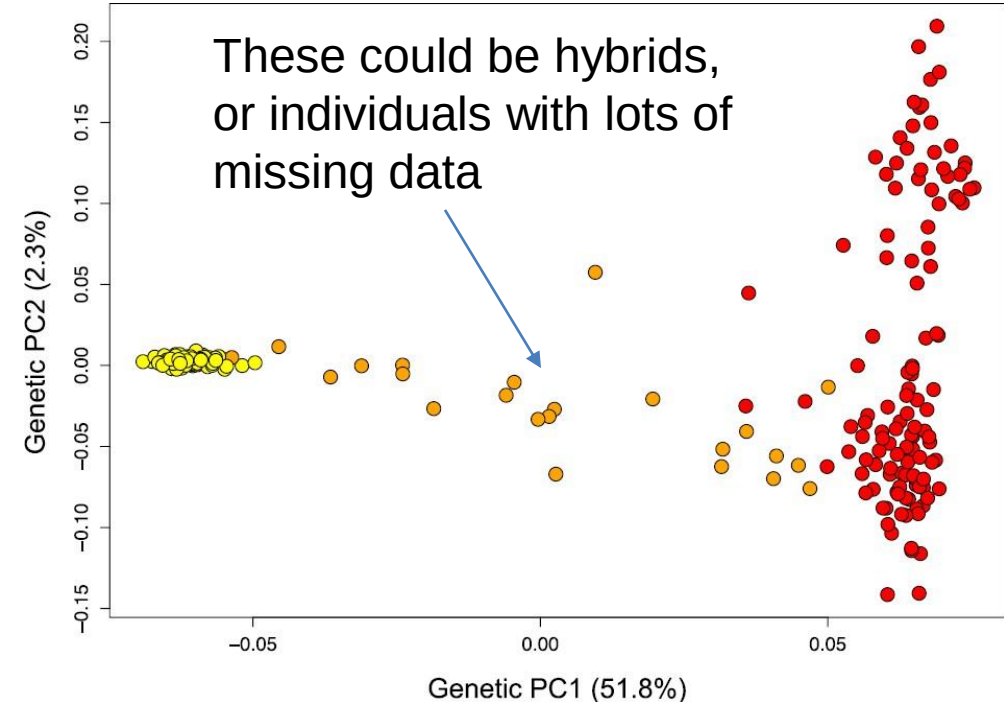
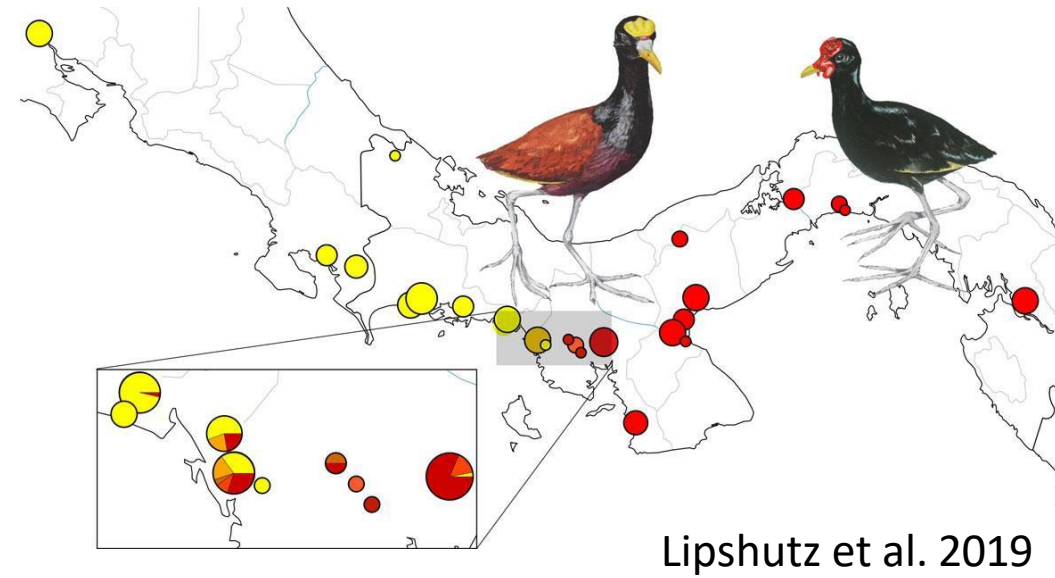
Why run a PCA?

- Are all individuals clustering together as expected? (e.g. no outlier or misidentified individuals)
- Testing if individuals with different colour, morphology or geographic origin cluster separately.
- Are any individuals genetically intermediate?



Why run a PCA?

- Are all individuals clustering together as expected? (e.g. no outlier or misidentified individuals)
- Testing if individuals with different colour, morphology or geographic origin cluster separately.
- Are any individuals genetically intermediate?

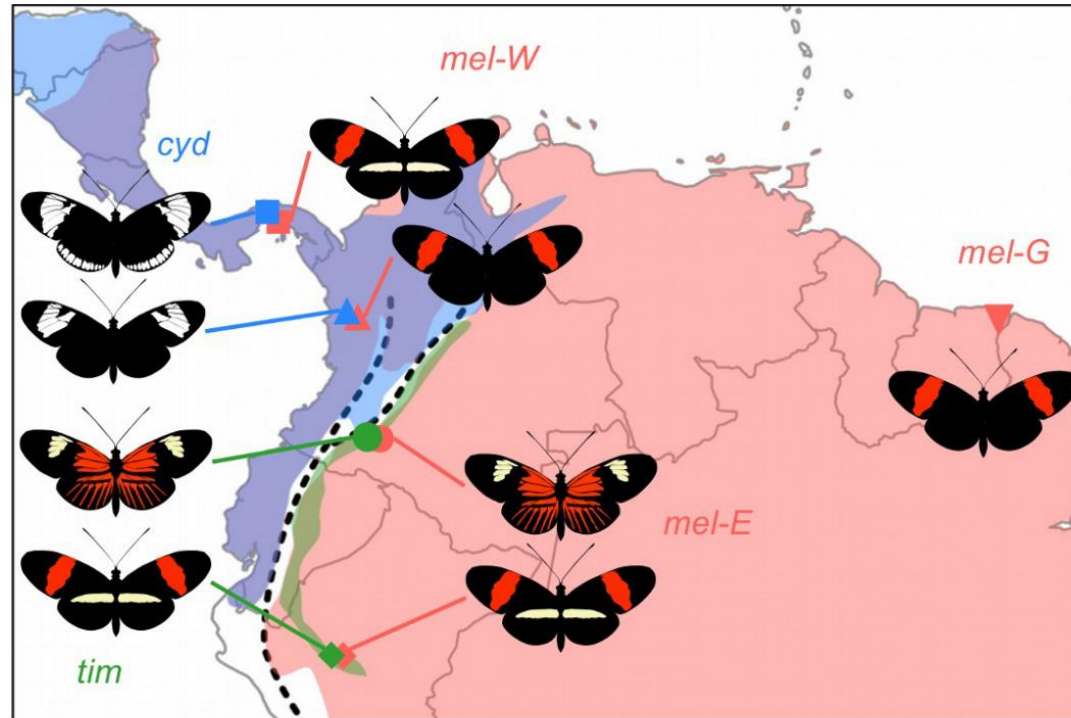


Assumptions of PCA

- All sites are independent
- For genomic data:
only SNPs (no insertions or deletions)
- Individuals are closely related and populations are more or less evenly represented.

If you have only few individuals each of divergent species
-> better use phylogenetic approaches

Dataset we will use for this tutorial



mel-W | ■ *H. melpomene rosina*
| ▲ *H. m. vulcanus*

mel-E | ● *H. m. malleti*
| ◆ *H. m. amaryllis*

mel-G | ▼ *H. m. melpomene*

cyd | ■ *H. cydno chioneus*
| ▲ *H. c. zelinde*

tim | ● *H. timareta florencea*
| ◆ *H. t. thelxinoe*